



魅力统计

Fascinating Statistics

第四章 数据及可视化

Chapter 4 Data and Visualization

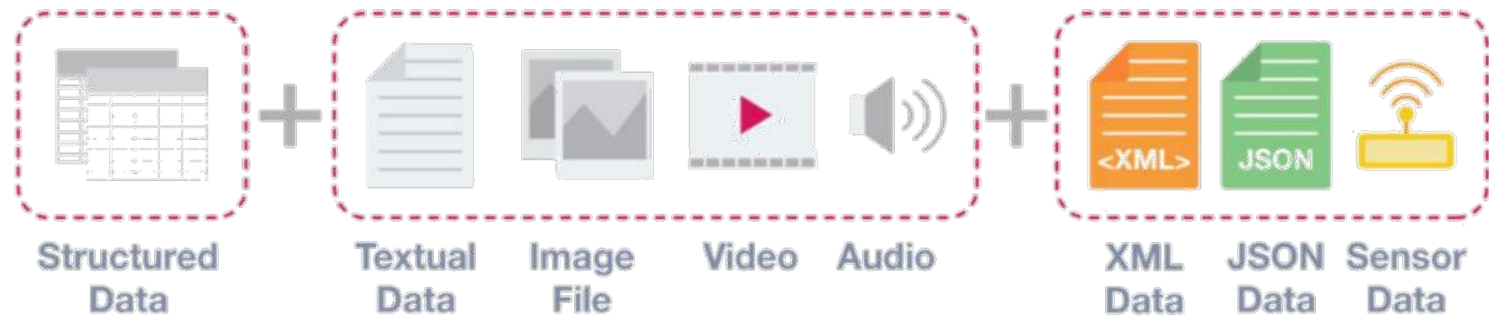
Chapter 4 Data and Visualization

- 4.1 Data Collection
- 4.2 Data Visualization



4.1 Data Collection

- Statistics studies how to measure, collect, organize, summarize, and analyze data to help people correctly understand the regularity of the objective world.
- Data collection is the foundation for subsequent organization, summarization, and analysis, so learning about data is essential.
- In a broad sense, data refers to collections of numbers, text, images, sound, etc., gathered from different sources, which is now referred to as **multi-modal data** (多模态数据) in the field of artificial intelligence.
- In a narrow sense, data mainly refers to **structured data** (结构化数据) composed of textual symbols representing categories and numbers.



4.1 Data Collection

- Data can be categorized based on its structure into:
 - **Structured data (结构化数据)**: Highly organized and neatly formatted data that can be easily stored in 2D tables. Each row represents information about an individual (also called an instance, case, or record), and each column represents a **feature** (特征, also called an attribute, variable, or field).
 - **Unstructured data (非结构化数据)**: Irregular data with no predefined data model, not easily represented in 2D logical tables. Examples include text, images, audio, and video.
 - **Semi-structured data (半结构化数据)**: A form of structured data that does not conform to the structural constraints of a relational database model but contains markers to separate semantic elements and hierarchically organize records and fields. The structure and content are intertwined without clear separation, such as HTML, JSON, and XML documents.

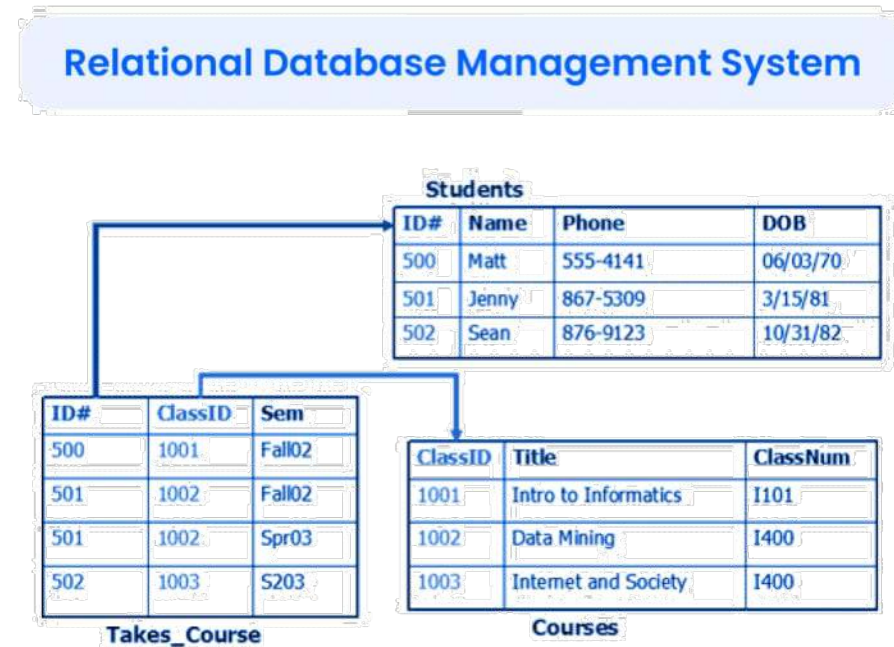
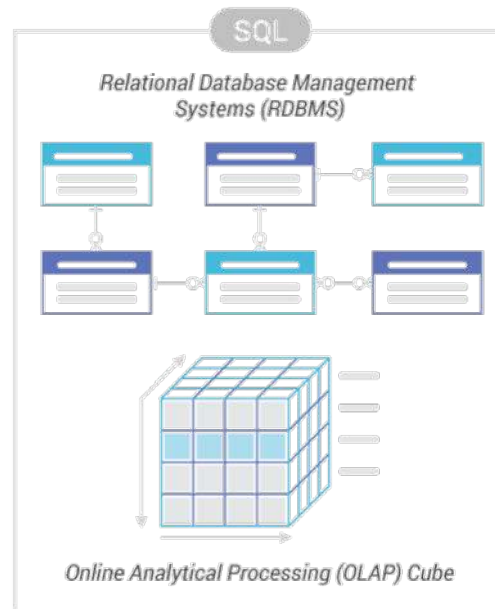
A screenshot of a web browser or code editor showing an HTML file named 'index.html'. The code defines a table with two columns: 'Version' and 'Date'. The table has a header row and two data rows. The first data row shows '2020.2' for Version and 'July, 2020' for Date. The second data row is partially visible, showing '2020.2' for Version.

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head...>
9   <body>
10    <div class="container">
11      <table class="table">
12        <thead>
13          <tr>
14            <th class="align" scope="col">Version</th>
15            <th class="align" scope="col">Date</th>
16          </tr>
17        </thead>
18        <tbody>
19          <tr>
20            <th class="align" scope="row">2020.2</th>
21            <th class="align" scope="row">July, 2020</th>
```



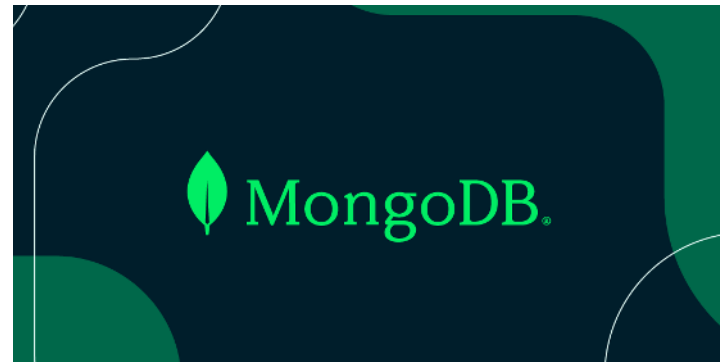
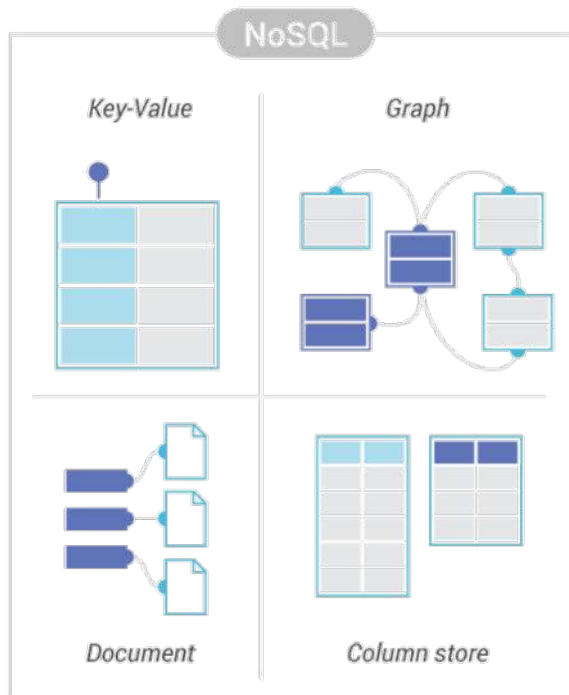
4.1 Data Collection

- Different types of data require distinct storage and processing methods:
 - Structured data is typically stored in **relational database management system** (关系型数据库, **RDBMS**), where data is organized across multiple tables with predefined logical relationships.
 - Well-known RDBMS include MySQL、PostgreSQL、Microsoft SQL Server、Oracle Database.
 - The programming language for operating RDBMS is **SQL** (Structured Query Language, 结构化查询语言), developed by IBM in 1974.



4.1 Data Collection

- Different types of data require distinct storage and processing methods:
 - Unstructured data is typically stored in **NoSQL (Not Only SQL) databases (非关系型数据库)**, which do not store data in rule-based tabular formats but instead retain it as separate, unconnected files.
 - Well-known NoSQL databases include MongoDB, Redis, Neo4j, etc.
 - NoSQL databases lack a standard language, and there are significant differences between them.



redis



4.1 Data Collection

- Unstructured data accounts for a far larger volume than structured data, representing over 80% of all enterprise data, and this proportion continues to grow.
- As statisticians or data scientists, we generally do not need expertise in how data is stored — this falls under the domain of data engineers. Our focus is on querying databases and extracting the data required for subsequent analysis.
- Analyses of structured data (e.g., statistical methods, models, and machine learning algorithms) are already well-established. However, advancements in deep learning have unlocked breakthroughs in extracting value from unstructured data—topics we will explore in detail later.
- Before diving into these topics, let's first understand how data is collected.
- Most apps we use daily comprehensively collect user information. **Data is not gathered to address a specific research question but rather stored first, with values explored afterward.**
- **UK Biobank** follows a similar philosophy. It is a large-scale biomedical database containing lifestyle habits, health records, biological samples, medical imaging, whole-genome sequencing data, and more from 500,000 UK participants.



4.1 Data Collection

UK Biobank participants have generously provided a very wide range of information about their health and well-being since recruitment began in 2006. This has been added to in the following ways:

- **Imaging:** Brain, heart and full body MR imaging, plus full body DEXA scan of the bones and joints and an ultrasound of the carotid arteries. The goal is to image 100,000 participants, and to invite participants back for a repeat scan some years later.
- **Genetics:** Whole genome sequencing for all 500,000 participants, whole exome sequencing for 470,000 participants, genotyping (800,000 genome-wide variants and imputation to 90 million variants).
- **Health linkages:** Linkage to a wide range of electronic health-related records, including death, cancer, hospital admissions and primary care records.
- **Biomarkers:** Data on more than 30 key biochemistry markers from all participants, taken from samples collected at recruitment and the first repeat assessment.
- **Activity monitor:** Physical activity data over a 7-day period collected via a wrist-worn activity monitor for 100,000 participants plus a seasonal follow-up on a subset.
- **Online questionnaires:** Data on a range of exposures and health outcomes that are difficult to assess via routine health records, including diet, food preferences, work history, pain, cognitive function, digestive health and mental health.
- **Repeat baseline assessments:** A full baseline assessment is undertaken during the imaging assessment of 100,000 participants.
- **Samples:** Blood & urine was collected from all participants, and saliva for 100,000.



4.1 Data Collection

THE WHITE HOUSE



FEBRUARY 28, 2024

Executive Order on Preventing Access to Americans' Bulk Sensitive Personal Data and United States Government- Related Data by Countries of Concern



BRIEFING ROOM

PRESIDENTIAL ACTIONS

Sec. 6. Assessing Risks Associated with Human 'omic Data Within 120 days of the date of this order, the APNSA, the Assistant to the President and Director of the Domestic Policy Council, the Director of the Office of Science and Technology Policy, and the Director of the Office of Pandemic Preparedness and Response Policy, in consultation with the Secretary of State, the Secretary of Defense, the Secretary of Health and Human Services, the Secretary of Veterans Affairs, the Director of the National Science Foundation, the Director of National Intelligence, and the Director of the Federal Bureau of Investigation, shall submit a report to the President, through the APNSA, assessing the risks and benefits of regulating transactions involving types of human 'omic data other than human genomic data, such as human proteomic data, human epigenomic data, and human metabolomic data, and recommending the extent to which such transactions should be regulated pursuant to section 2 of this order. This report and recommendation shall consider the risks to United States persons and national security, as well as the economic and scientific costs of regulating transactions that provide countries of concern or covered persons access to these data types.



4.1 Data Collection

PRESS RELEASE

Justice Department Issues Final Rule Addressing Threat Posed by Foreign Adversaries' Access to Americans' Sensitive Personal Data

Friday, December 27, 2024

For Immediate Release

Office of Public Affairs

Final Rule Implements Executive Order to Prevent Access to Americans' Bulk Sensitive Personal Data and U.S. Government-Related Data by Russia, Iran, China and Other Countries of Concern

Note: [View the fact sheet here:](#)

WASHINGTON – Today, the Justice Department issued a comprehensive [final rule](#) carrying out Executive Order (E.O.) 14117 “Preventing Access to Americans’ Bulk Sensitive Personal Data and United States Government-Related Data by Countries of Concern.” The E.O. charged the Justice Department with establishing and implementing a new regulatory program to address the urgent and extraordinary national security threat posed by the continuing efforts of countries of concern (and covered persons that they can leverage) to access and exploit Americans’ bulk sensitive personal data and certain U.S. Government-related data. The Final Rule will take effect 90 days from the date of the Final Rule’s publication, with certain affirmative due diligence, reporting, and auditing requirements taking effect 270 days after publication.



4.1 Data Collection

- From a positive perspective, we should seize this opportunity to advance the development of China's own biological databases.
- China possesses unparalleled advantages in biological data resources:
 - Massive population and clinical data.
 - Rich human genetic and pathological resources.
 - Abundant plant and animal genetic resources.
- Current progress in database development:
 - China has established the **National Genomics Data Center** (国家基因组科学数据中心) and initiated preliminary collaborations with selected European and American databases.
 - Many universities and research institutes have also built decentralized databases for specialized fields.



面向我国人口健康和社会可持续发展的重大战略需求，建立生命与健康大数据汇交存储、安全管理、开放共享与整合挖掘研究体系，研发大数据前沿交叉与转化应用的新方法和新技术，建成支撑我国生命科学发展、国际领先的基因组科学数据中心。



4.1 Data Collection

- Existing gaps:
 - **Infrastructure and integration:** we lag behind the U.S., Europe, and Japan in database infrastructure, integration, management, and search/cross-reference capabilities.
 - **Data quality and usability:** challenges remain in data authority, completeness, user-friendly interfaces, data accessibility, and intelligent cross-dataset linkages.
- Building and refining China's biological database will unlock immense opportunities for its biotech industry.
- By addressing current limitations and leveraging our resources, we can transform biological data into a strategic asset, positioning itself as a global leader in data-driven biomedical innovation.



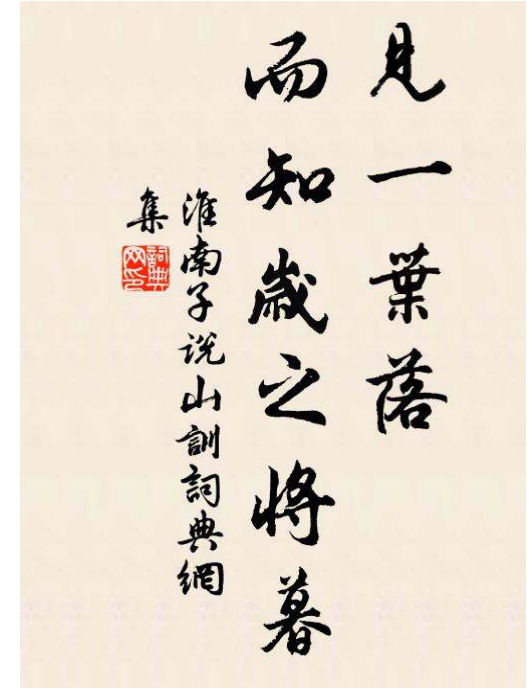
4.1 Data Collection

- In many cases, we have a specific research question that requires data collection tailored to address it. However, it is often infeasible to collect data from all subjects of interest.
 - The entire group of subjects of interest is termed the **population** (总体), and its constituent elements are called **individuals** (个体).
 - In statistics, a **population can be conceptualized as a r.v. X** , our goal is to select a subset of individuals from this population and obtain data to infer the probability distribution of X .
 - The process of selecting individuals from a population is referred to as **sampling** (抽样).
 - Sampling from a population can be conceptualized as sampling from the probability distribution of X .
 - Before observation or experiment, the selected individuals are treated as random variables $X_1, X_2, \dots, X_n$, called a **sample** (样本) with **sample size** (样本容量) n .
 - After observation or experiment, the data obtained $(x_1, x_2, \dots, x_n)$ are termed the **sample values** (样本值), or sample observed values (样本观察值).



4.1 Data Collection

- The idea that a local perspective can reveal universal truths has long been recognized, there is a famous text: "From the small, we discern the large; seeing a single leaf fall, we know that the year is coming to an end; observing ice in a bottle, we understand the coldness of the world."
- However, sometimes a local perspective does not provide a complete understanding of the whole situation. For example, there is a proverb: "peering at a leopard through a tube, one sees only spots."
- **Question:** under what circumstances can conclusions drawn from a sample be generalized to the entire population?
- This requires special attention during sampling to ensure that **the samples drawn are sufficiently representative (样本要有足够的代表性)**.



4.1 Data Collection

Example 4.1

Public opinion polling in U.S. presidential elections

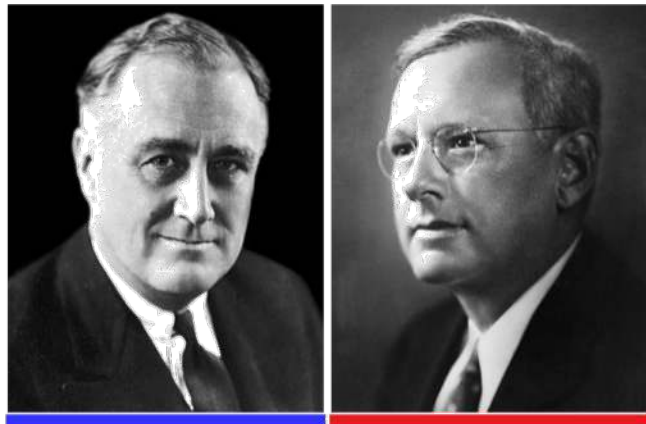
- The United States has numerous polling agencies, all striving for accuracy in their polls, as it is crucial for maintaining their reputation and even their survival and growth.
- From 1916 to 1932, the dominant player in polling was *The Literary Digest*. It accurately predicted the results of 5 consecutive presidential elections, earning widespread trust and acclaim.
- In 1932, *The Literary Digest* successfully predicted the election outcome, ushering the era of Roosevelt's New Deal.
- In the 1936 election, the two candidates were Roosevelt (Democratic) and Landon (Republican). Most polling organizations, media outlets, and political observers forecasted Roosevelt's victory.
- However, *The Literary Digest* offered a different prediction. It conducted a large-scale survey, mailing out a staggering 10 million questionnaires and receiving about 2.3 million responses.
- The results indicated that 57% of respondents supported Landon.



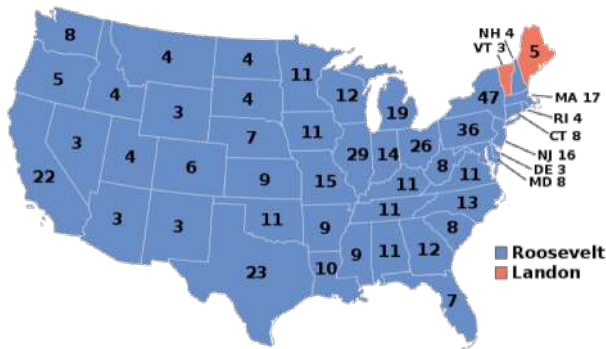
4.1 Data Collection

Example 5.2

- Due to the significant influence of *The Literary Digest*, many believed that Landon would become the 33rd President of the U.S.
- However, it turned out that Roosevelt achieved a landslide victory, winning by the largest margin in the history of U.S. presidential elections.
- This grave error destroyed *The Literary Digest's* credibility, leading to its demise within 18 months of the election.
- In contrast, a relatively unknown company, Gallup, correctly predicted the outcome with a sample of only 50,000 respondents, propelling it to fame.
- What went wrong with *The Literary Digest's* polling?



Nominee	Franklin D. Roosevelt	Alf Landon
Party	Democratic	Republican
Home state	New York	Kansas
Electoral vote	523	8
States carried	46	2
Popular vote	27,747,636	16,679,543
Percentage	60.8%	36.5%



4.1 Data Collection

Solution

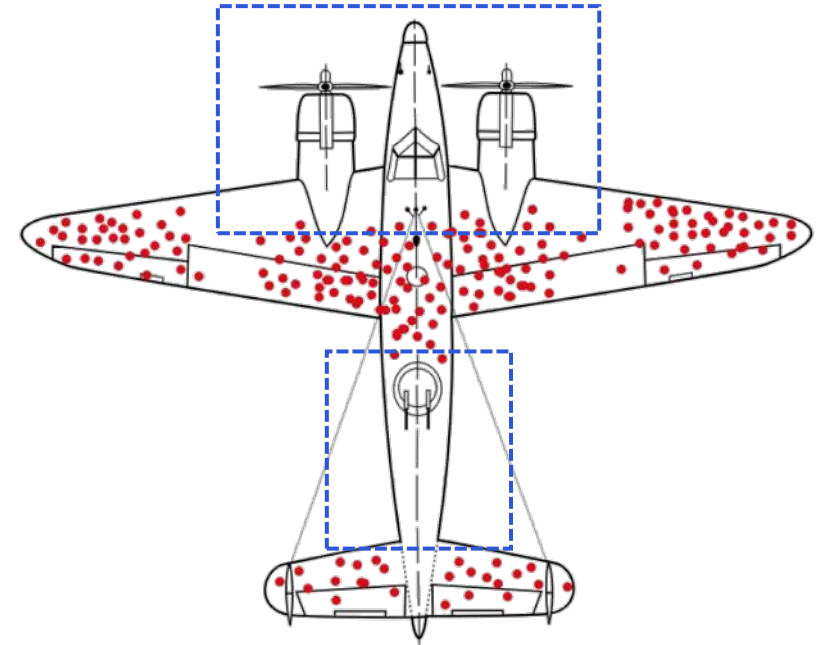
- Years later, a 1976 article in *The American Statistician* identified systematic sampling bias in *The Literary Digest's* polling.
- Among the 10 million questionnaires mailed, the 2.3 million responses reflected a **non-response bias** (无回应偏差): voters opposing Roosevelt, though a minority in the population, were more motivated to respond to the survey.
- Further analysis in 1988 revealed an additional source of systematic sampling bias. The Literary Digest selected its 10 million recipients from telephone directories and club membership lists.
- In 1936, telephone ownership was limited, predominantly to wealthier, politically conservative, and Republican-leaning households. This introduced **selection bias** (选择偏差).
- Both forms of bias lead to the same issue: the **sample was not sufficiently representative of the population**.



4.1 Data Collection

Example 4.2

- During World War II, the United States sought to reinforce specific areas of its bombers with additional armor.
- Analysts examined returning bombers and mapped the bullet holes and damage.
- Based on this analysis, they concluded that adding armor to the tail, fuselage, and wings would increase the pilots' chances of survival.
- What is your perspective on this approach?



4.1 Data Collection

Solution

- At the time, a young statistician named Abraham Wald pointed out that this would be a tragic mistake.
- He realized that the returning bombers had damage they could survive, while the others had suffered catastrophic damage and never made it back.
- Therefore, his conclusion was to add armor to the "undamaged" areas of the returning bombers.
- Based on his insight, the military reinforced these components, significantly enhancing the safety of crews in combat and saving many lives.
- Wald identified a phenomenon known as **survivorship bias (幸存者偏差)**: focusing on individuals who pass a selection process (i.e., survivors) while ignoring those who do not, leading to erroneous conclusions.
- Survivorship bias can be understood as a form of non-response bias or selection bias mentioned earlier, fundamentally stemming from the lack of representativeness in the observed sample.



4.1 Data Collection

- The examples above demonstrate that to derive meaningful conclusions, it is imperative to rigorously minimize potential biases during the data collection process.
- Additionally, one should maintain a healthy degree of critical thinking (合理的批判性思维) when evaluating conclusions drawn from others' data analyses.



丁香园 >



成精各种猫 辽宁 3天前

回复 138

有没有一种可能，这样的人不止一个，有些人因为出现了不良反应，不敢继续这样做了？



精神状态稳中... 广东 3天前

回复 137

所以中科院博士也提醒这个样本局限性很大



糖芋苗大肉包... 江苏 3天前

回复 17

可能他个体可以承载这么多的疫苗，而别人未必。真的是个体差异



梦寰拉萨 北京 2天前

回复 17

回复 糖芋苗大肉包（都不能吃）：所以科研人员在拿不到数据的时候，也怀疑他始终没被病毒感染不一定是高频接种的结果，也可能是他个体素质😂😂这就是个彪人，百毒不侵啊，谁来都不好使



陈明苍 上海 3天前

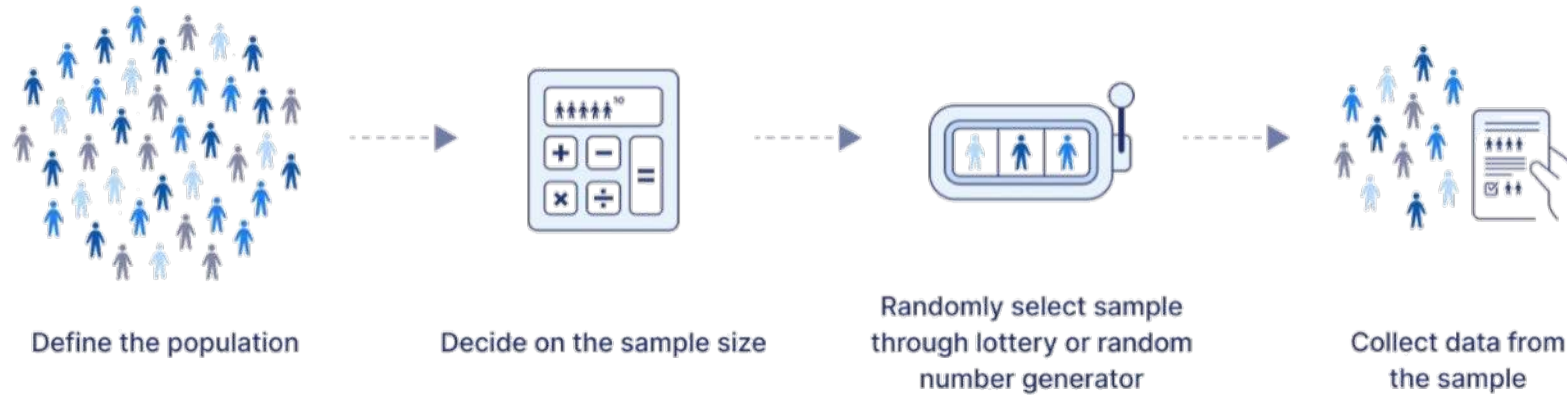
回复 134

首先这是个例，不具有普遍性。此外，这大爷没事不代表别人也没事，极有可能有事人的早就gg了，无从查起了。因此，如此高密度接种疫苗的安全性绝非如此安全😄



4.1 Data Collection

- The simplest method to ensure representativeness is **simple random sampling** (简单随机抽样).

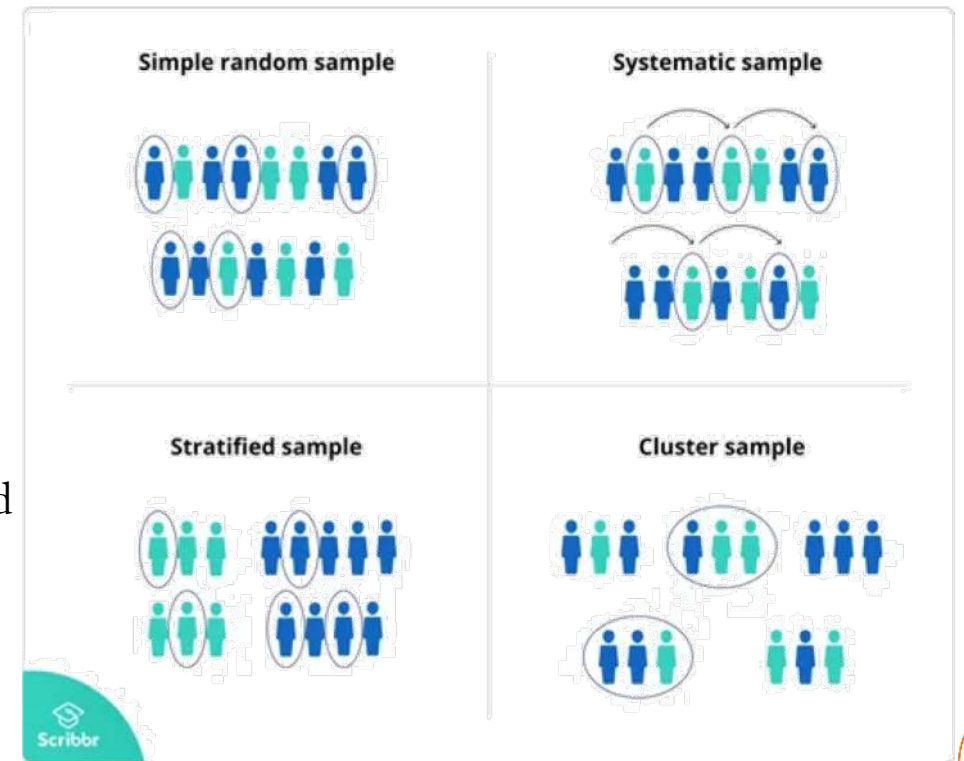


- The core of simple random sampling is to select samples in a completely random manner, ensuring that every individual in the population has an **equal probability of being chosen**.
- Expressed in statistical language, $X_1, \dots, X_n$ are independent, and follow the same distribution with X , i.e., $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} X$. This is a **fundamental assumption of many statistical methods**.
- However, it is difficult to strictly perform simple random sampling in reality, as it requires you to number each individual in the population and then draw individuals in a way that generates random numbers.



4.1 Data Collection

- Due to cost constraints and practical considerations, several commonly used random sampling methods are employed in real-world applications:
 - **Systematic sampling (系统抽样)**: Individuals are ordered based on some criterion and are selected at fixed intervals (e.g., every 10th individual).
 - **Stratified sampling (分层抽样)**: the population is divided into subgroups (strata) based on shared characteristics (e.g., age, gender). Individuals are drawn proportionally from each stratum.
 - **Cluster sampling (整群抽样)**: the population is divided into naturally occurring clusters (e.g., schools, neighborhoods, hospitals). A random set of clusters is chosen, and all individuals within the selected clusters are included in the sample.



Chapter 4 Data and Visualization

- 4.1 Data Collection
- 4.2 Data Visualization



4.2 Data Visualization

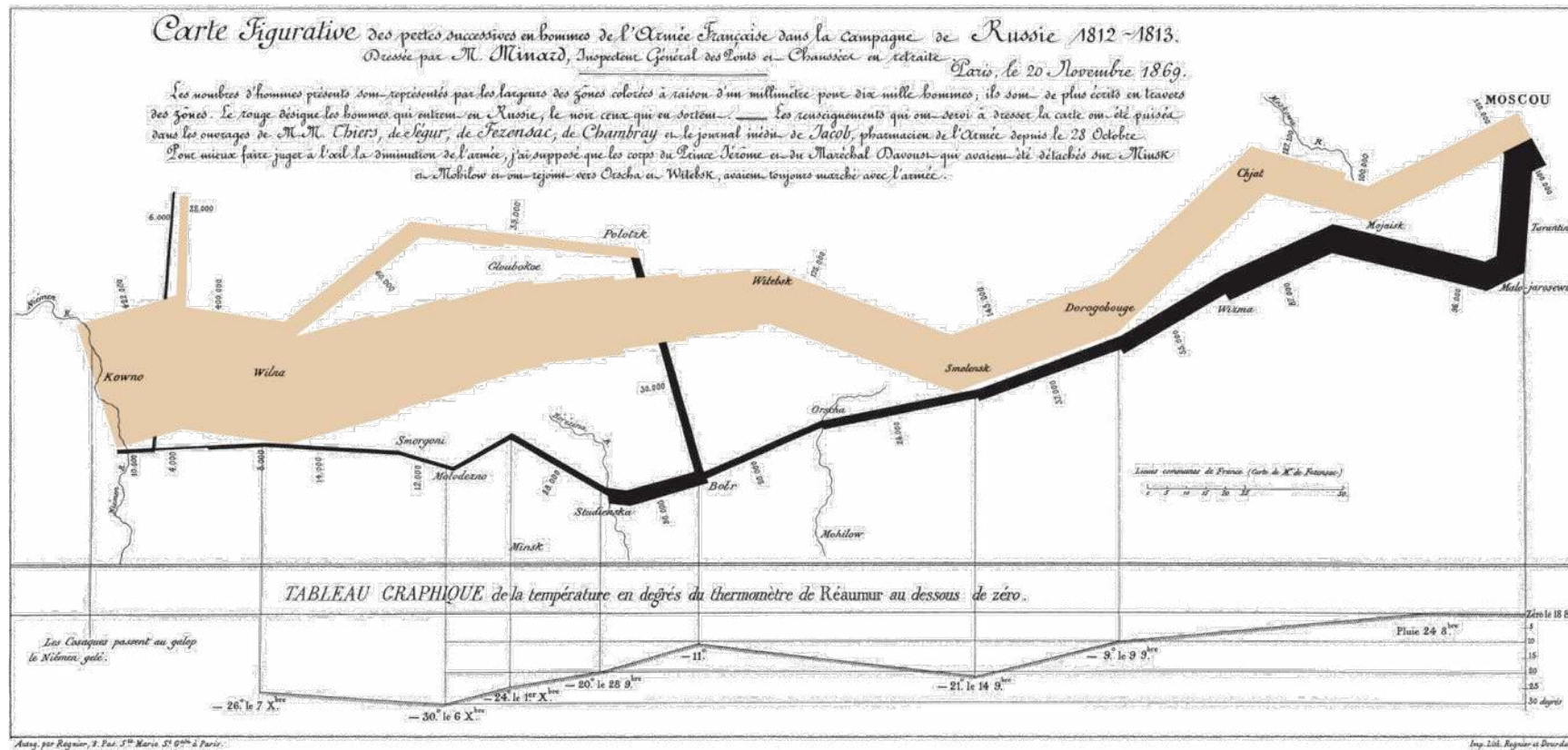
- The goal of data analysis is to extract value from data, and complexity is not a good thing.
- As the saying goes, “a picture is worth a thousand words”(一图胜千言), a simple graphic can reveal patterns in data without the need for extensive explanation.
- Data visualization acts as a universal language, breaking down barriers of technical jargon and linguistic differences to communicate insights across disciplines.
- The concept of visualizing data is ancient, predating computers by centuries. Techniques like maps, charts, and diagrams were used to represent information long before the digital era.
- In this section, we will
 - Explore historically significant examples and modern applications of data visualization.
 - Learn to create common statistical graphics using the [ggplot2 package in R](#).



4.2 Data Visualization

Example 4.3

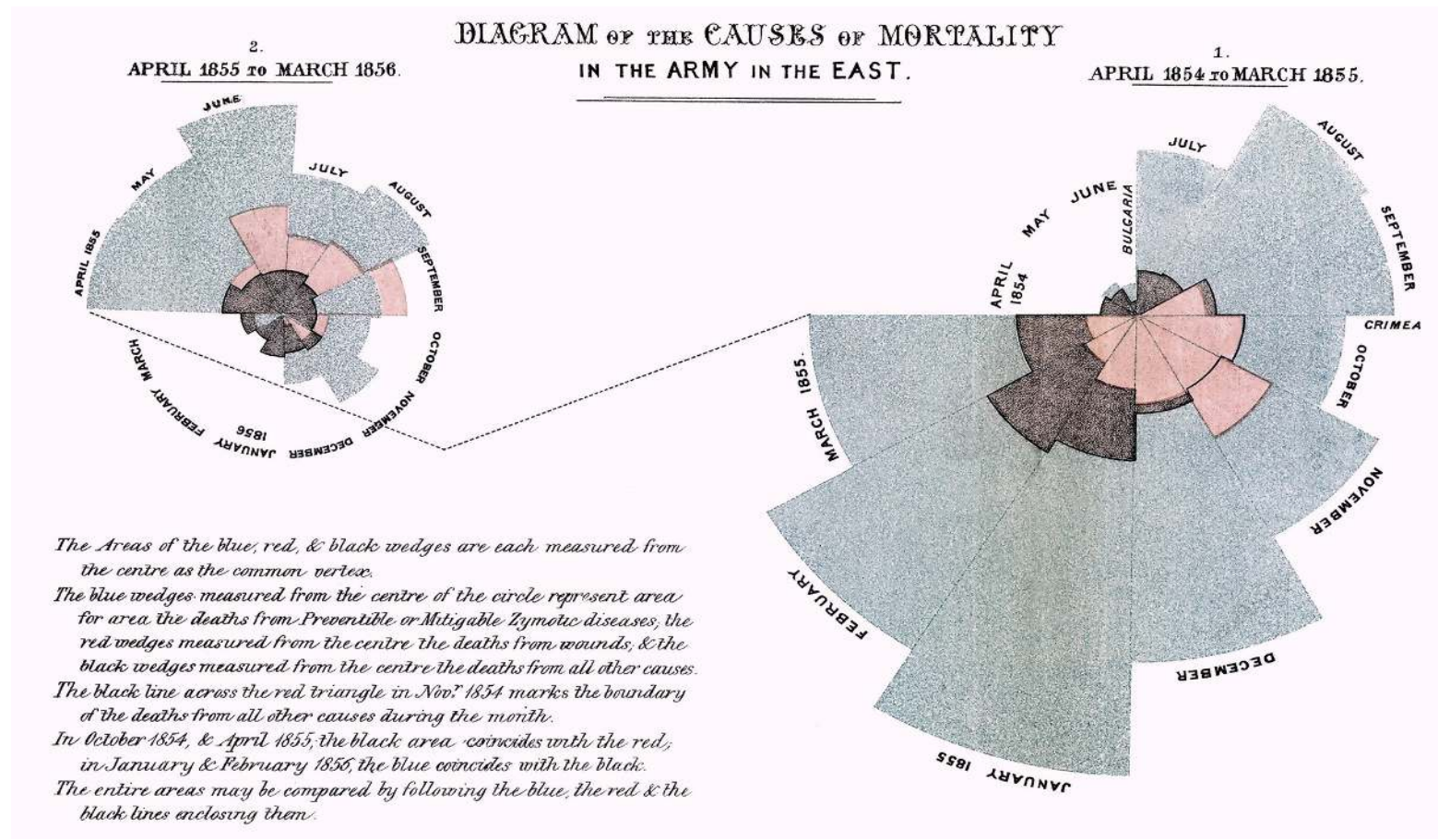
- Titled “The Continuous Losses of Napoleon’s Army During the 1812 – 1813 Russian Campaign,” by the civil engineer Charles Joseph Minard in 1869.



4.2 Data Visualization

Example 4.4

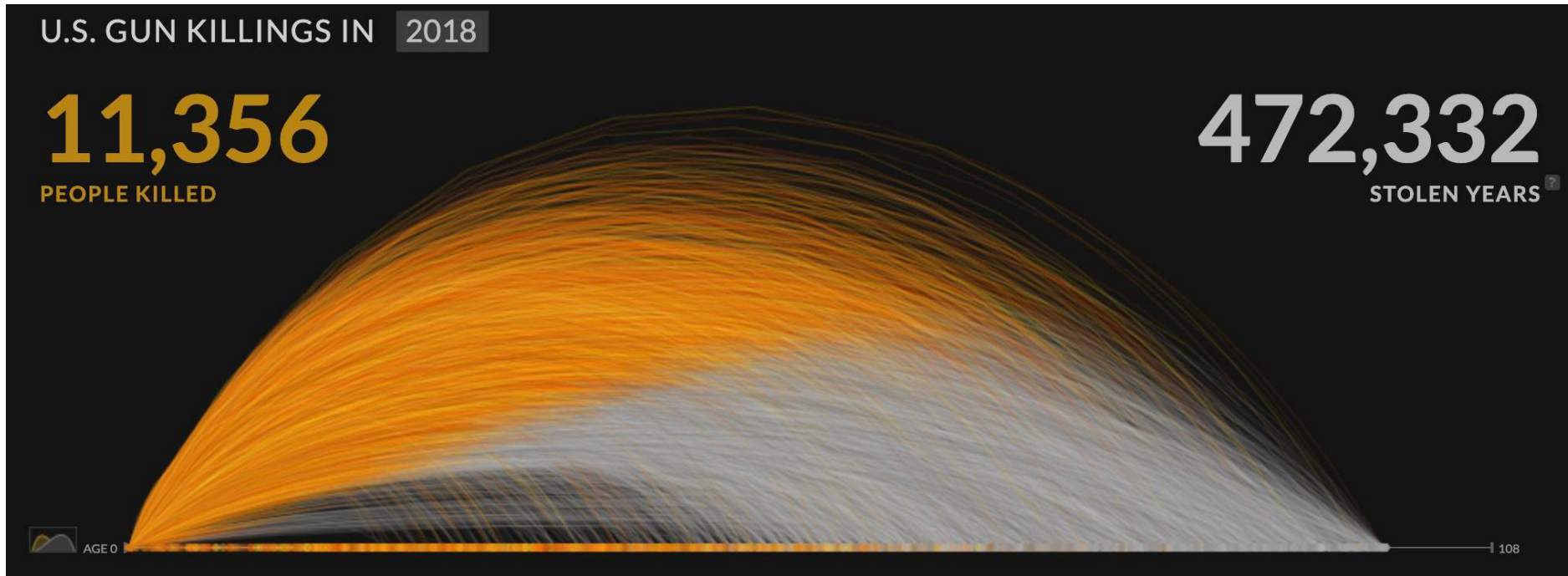
- Known as the “Nightingale Rose Chart,” by Florence Nightingale in 1858.



4.2 Data Visualization

Example 4.5

- U.S. Gun Killings in 2018 by Periscopeic.



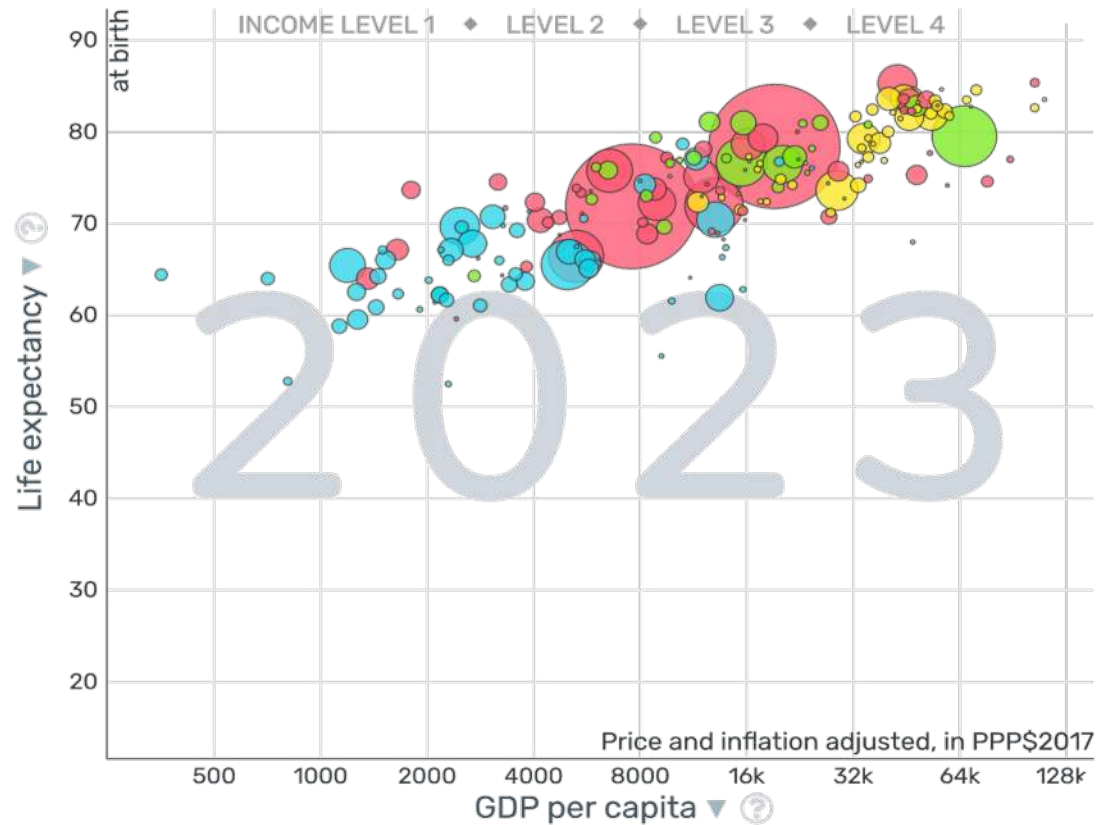
<https://guns.periscopeic.com>
<https://periscopeic.com/#!/>



4.2 Data Visualization

Example 4.6

- Gapminder by Hans Roseling.



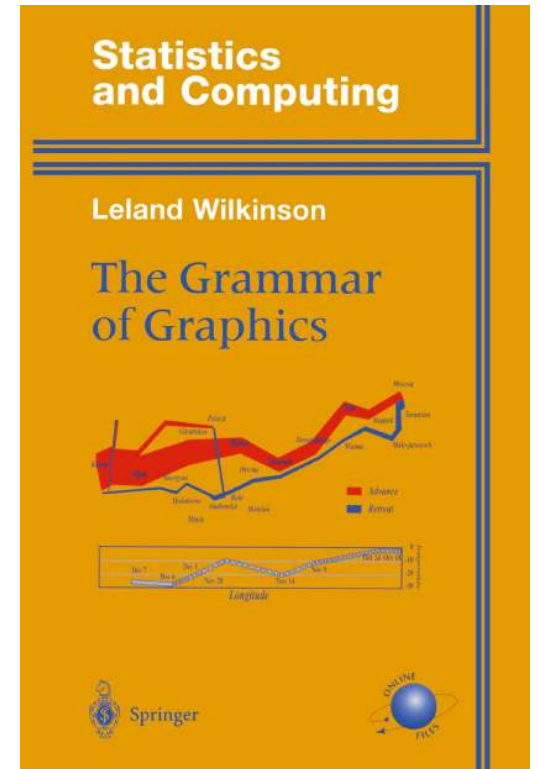
- Data visualization is often more creative and diverse than one might imagine.
- From business dashboards, public health visualizations to trend analysis in popular culture, it spans a wide spectrum of applications.
- To craft polished, high-quality visualizations, one needs not only strong analytical skills but also expertise in graphic design and storytelling.
- Core principles:
 - Less is more:** avoid excessive decoration; focus on the core message.
 - Tell a compelling story:** Guide the audience to discover patterns through visualization.

[https://www.gapminder.org/tools/#\\$chart-type=bubbles&url=v2](https://www.gapminder.org/tools/#$chart-type=bubbles&url=v2)



4.2 Data Visualization

- As statisticians or data scientists, it is essential to master at least one **data visualization tool**.
 - The Napoleonic campaign map and Nightingale's rose chart, mentioned earlier, were products of the hand-drawn era.
 - Today, however, we demand reusable technical implementations capable of generating visualizations dynamically for diverse datasets.
 - In 1999, **Leland Wilkinson (威尔金森)** pioneered a dedicated syntax for graphical programming and published *The Grammar of Graphics*.
 - This framework abandoned traditional point/line/area-based visualization methods.
 - Instead, it approached visualization through **data coordinate systems**, **relational mappings**, and **statistical transformations**, using a syntactic structure to describe data patterns while rendering them visually.
 - This methodology distinguished data analysts from mere “artists”, marking a revolutionary advancement—though it was not implemented at the time.



4.2 Data Visualization

- It wasn't until 200 that **Hadley Wickham** (哈德利·威克汉姆) developed the **ggplot2** package in R, finally realizing the vision of *The Grammar of Graphics*.
- Since then, programming languages like Python and MATLAB have adopted similar grammar-inspired visualization modules, reflecting ggplot2's enduring influence.
- Building on the conceptual framework of The Grammar of Graphics, ggplot2 empowers users to **define visualizations through composable elements**.
- In ggplot2, the elements of a plot include:
 - **Data (数据)**: a structured data table where each row represents an individual and each column represents a feature.
 - **Mapping (映射关系)**: a set of instructions defining how data map to visual properties (aesthetics), such as x/y-axis positions, color, shape, or size. These mappings act as a "dictionary" translating data into graphical elements.

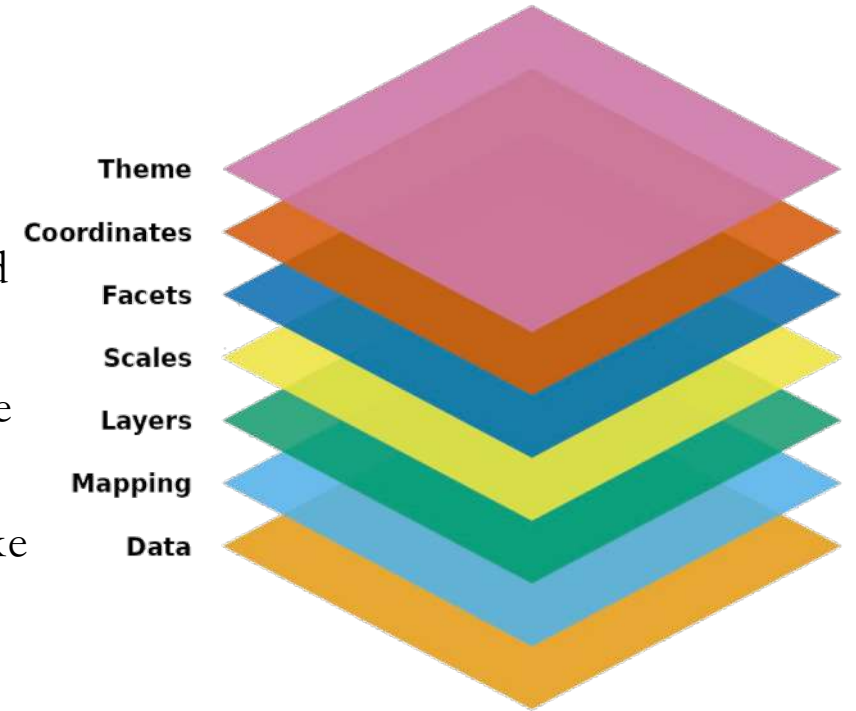


ggplot2官方网站: <https://ggplot2.tidyverse.org/>
ggplot2官方书籍: <https://ggplot2-book.org/>



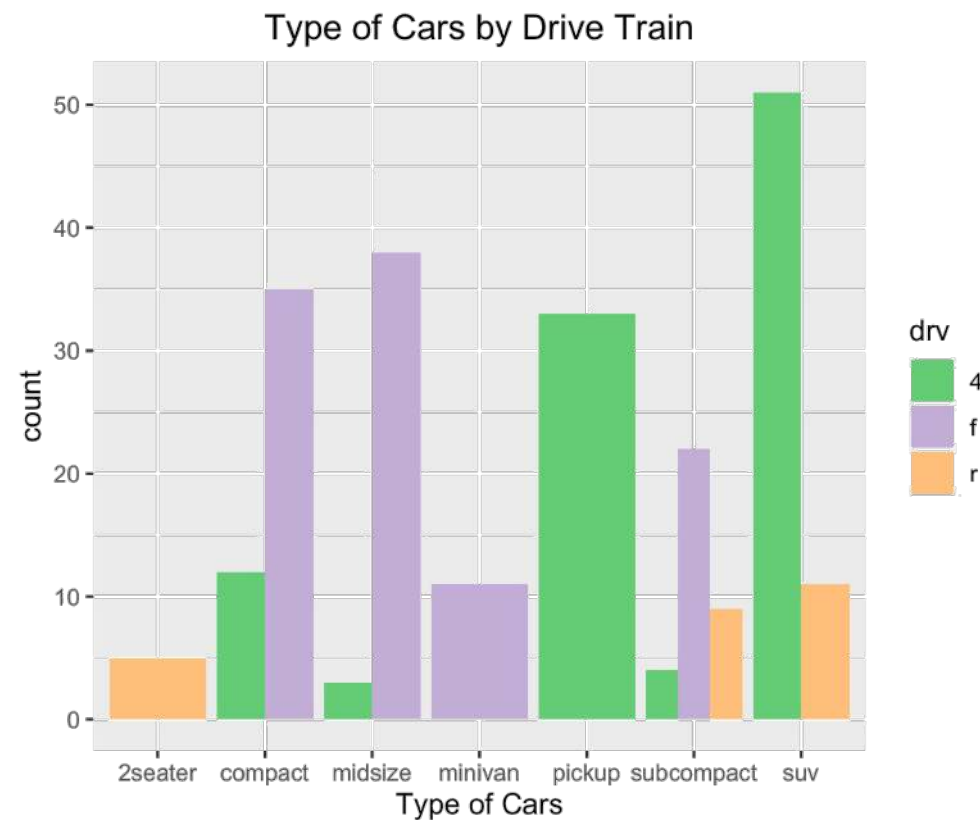
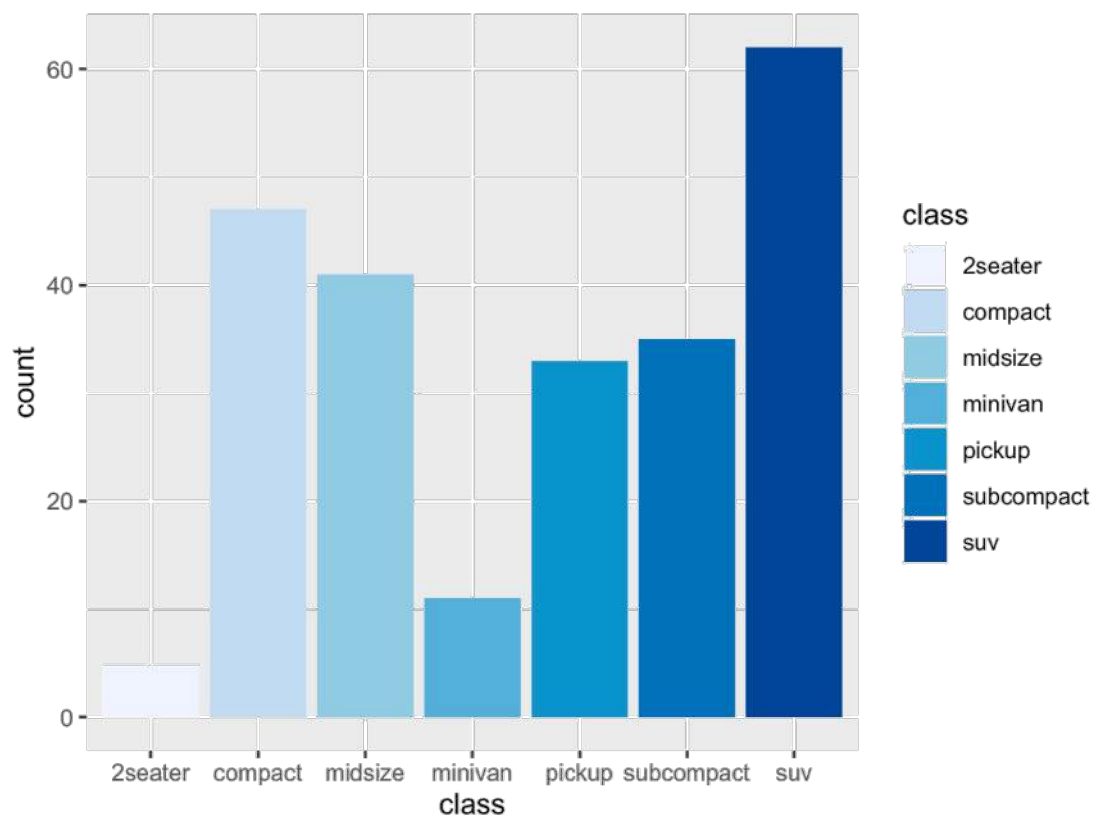
4.2 Data Visualization

- In ggplot2, the elements of a plot include :
 - **Layers (图层面板):** The core of the visualization, where mapped data is rendered into human-interpretable forms, consist of:
 - **Geometric objects** (geoms, 几何对象): visual primitives like points, lines, bars, or polygons;
 - **Statistical transformation** (stats, 统计变换): operations like binning, smoothing, or aggregation applied to the data.
 - **Plot details (图形细节):**
 - **Scales:** control how data values map to aesthetic attributes (e.g., color gradients, axis limits);
 - **Facets:** split the plot into subplots based on categorical features;
 - **Coordinates:** define the coordinate system (e.g., Cartesian, polar);
 - **Theme:** adjust non-data elements (e.g., titles, legends, fonts, background colors).
- By iteratively layering these components onto the base data, users can construct sophisticated, publication-quality statistical graphics with precision and elegance..



4.2 Data Visualization

- Other than the scatterplot (散点图) mentioned in Chapter 1, commonly used statistical graphs:
 - Bar plot / Bar chart (柱状图): for both categorical and numerical features.

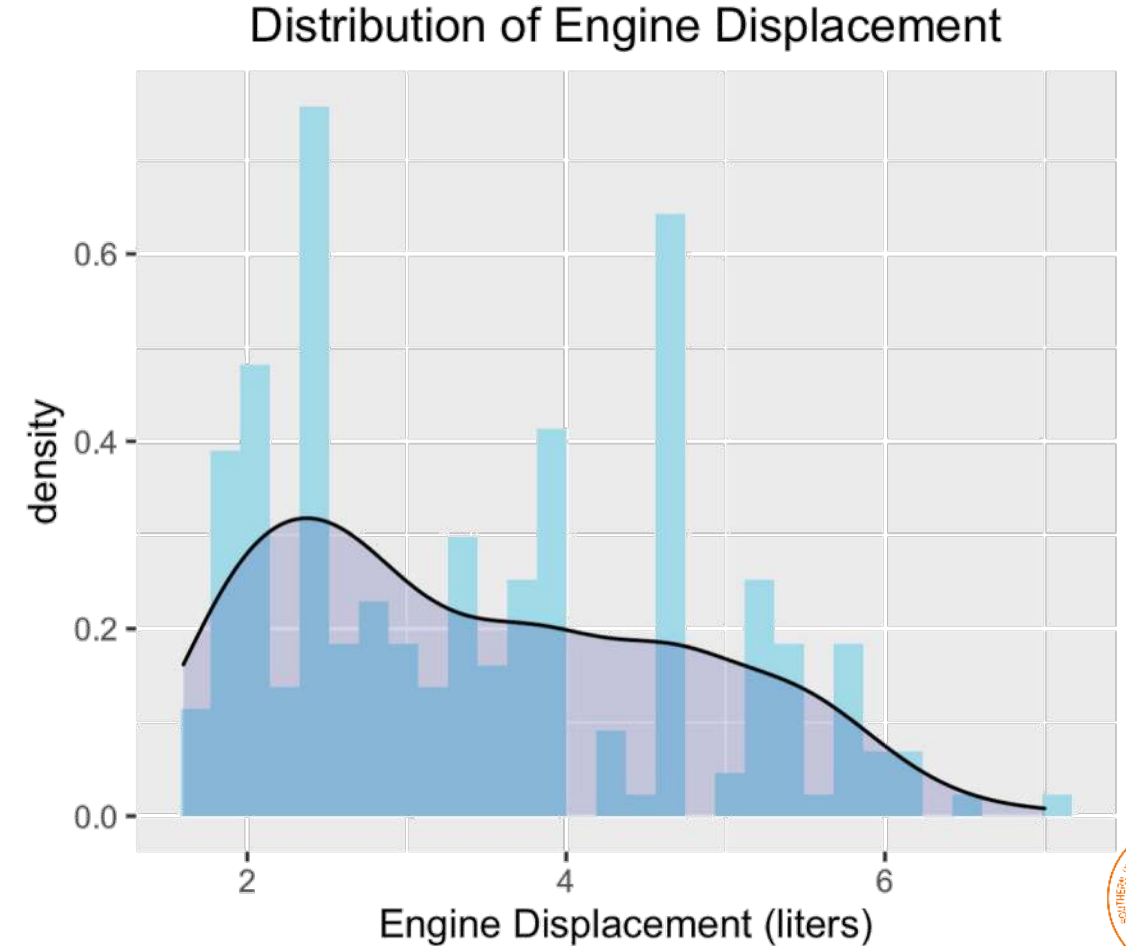
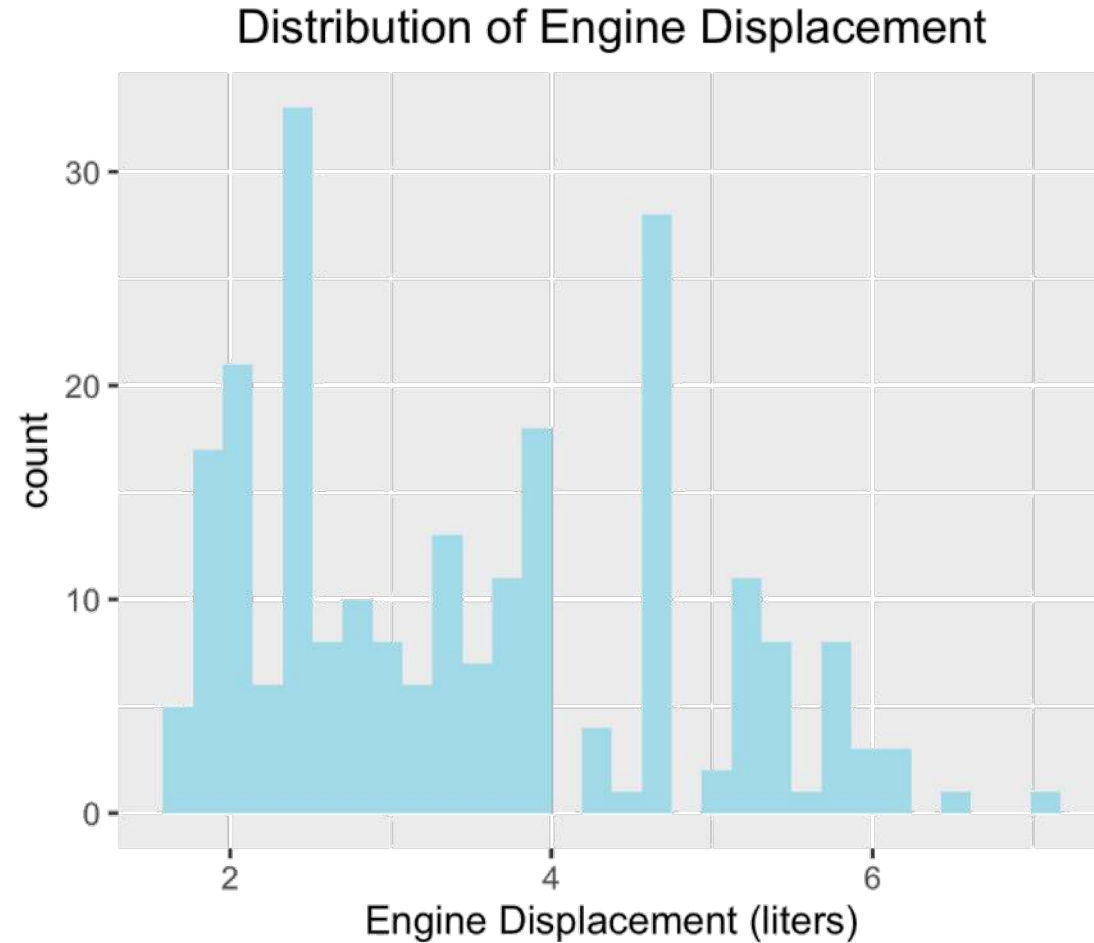


ggplot2 tutorial in Chinese: https://www.math.pku.edu.cn/teachers/lidf/docs/Rbook/html/_Rbook/ggplot2.html



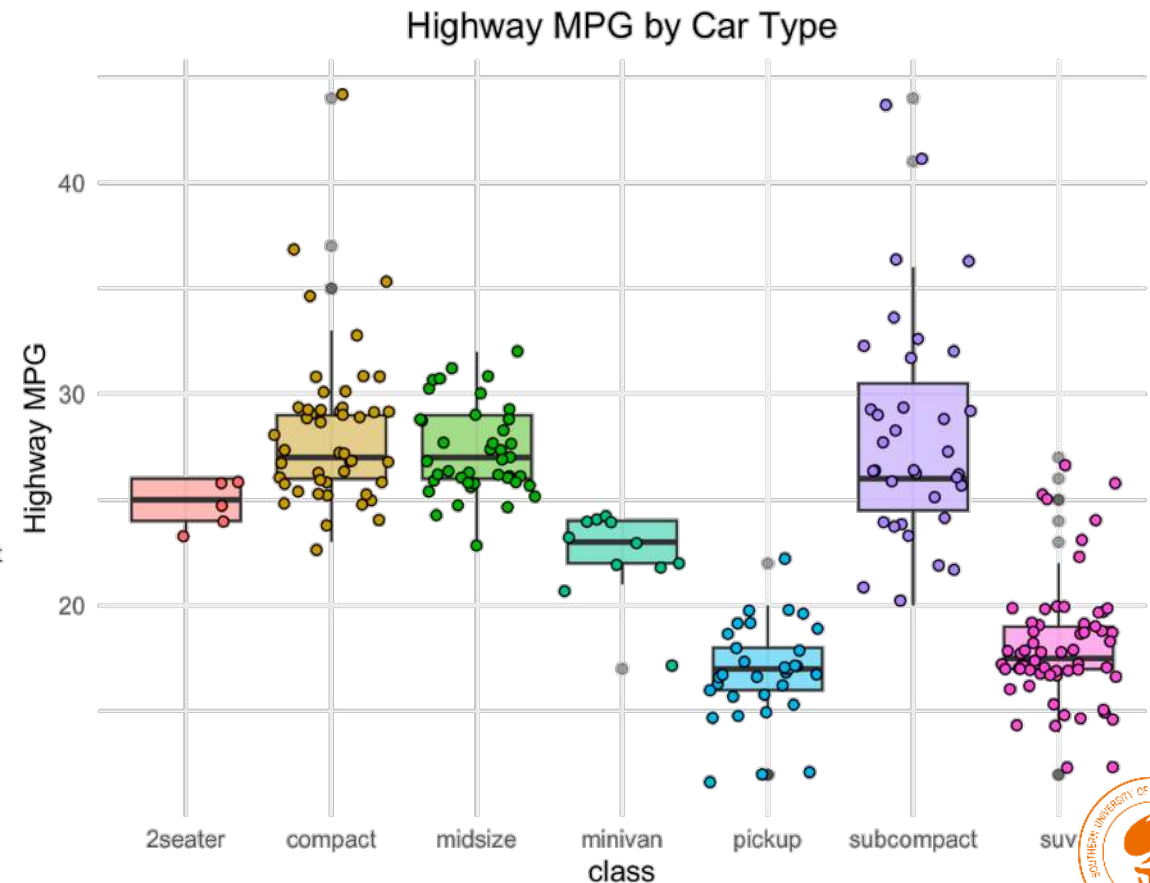
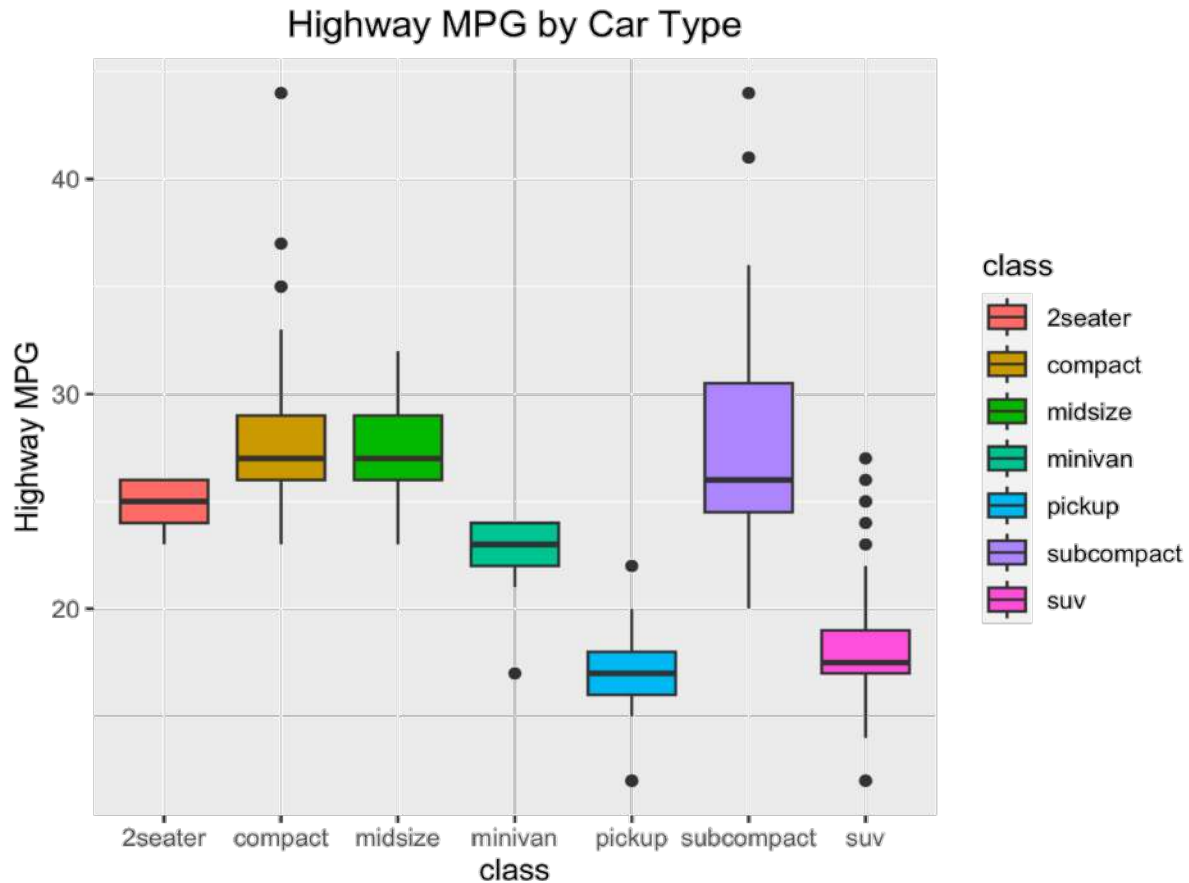
4.2 Data Visualization

- Histogram (直方图): for numerical features.



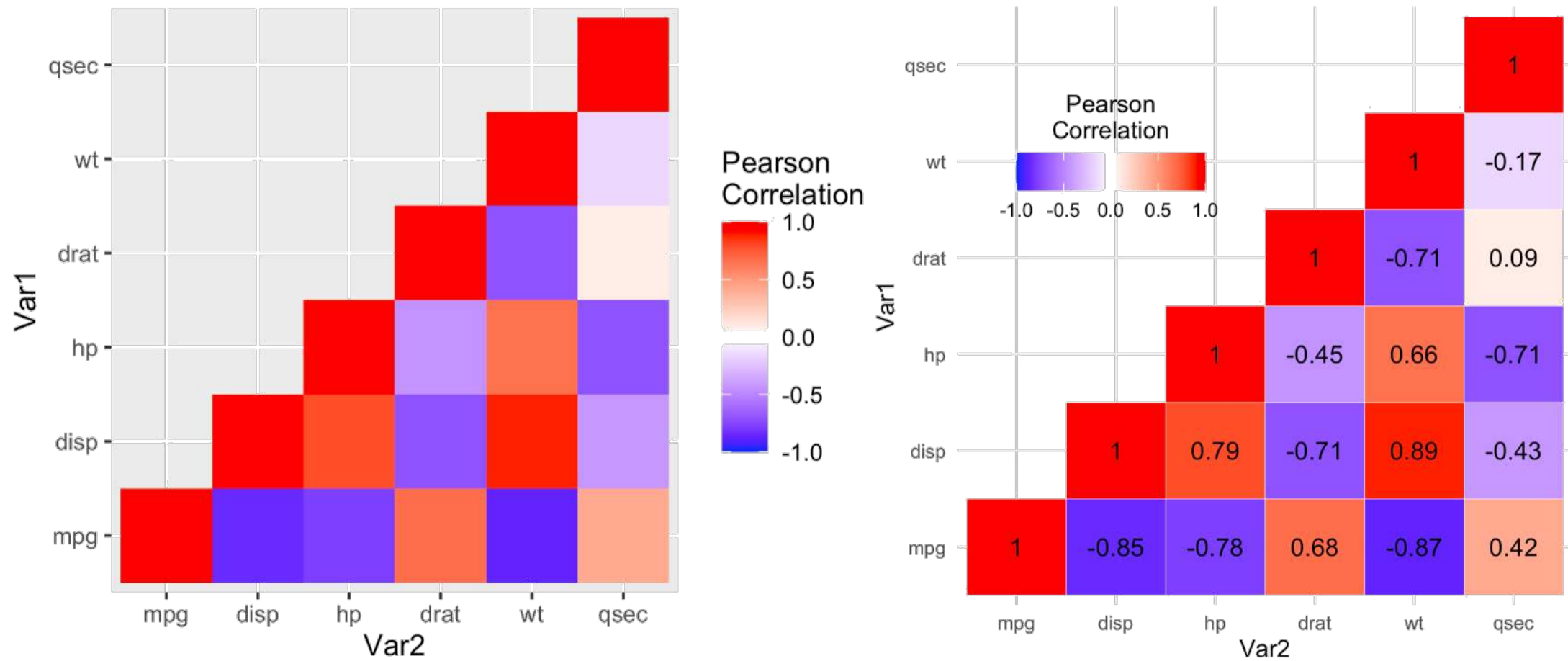
4.2 Data Visualization

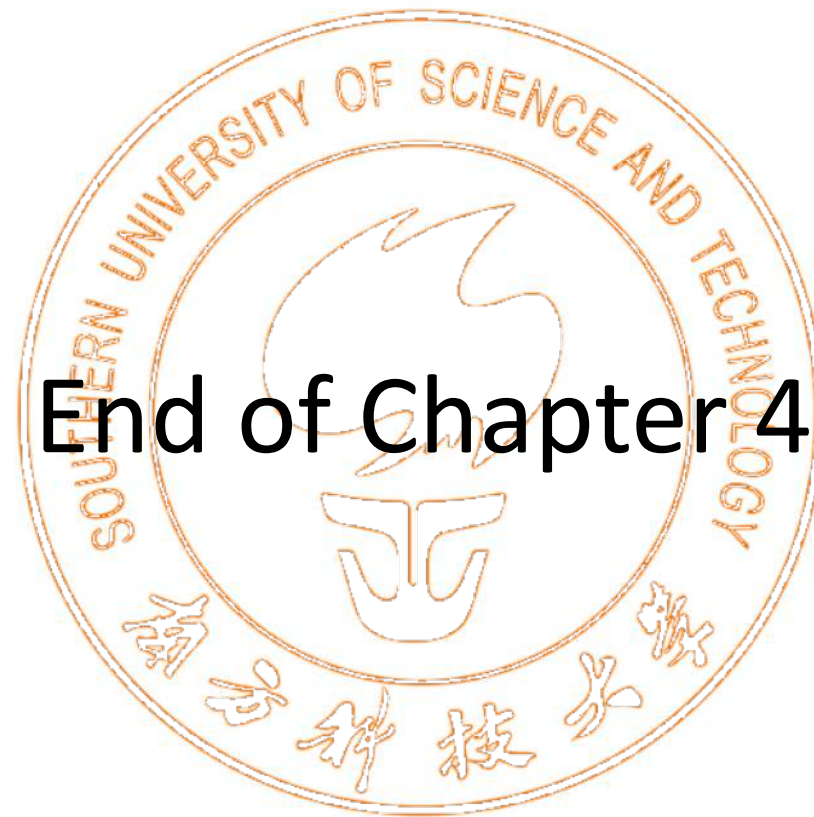
- **Box plot (箱线图)**: commonly used to compare the distribution of a numerical feature among different groups of individuals.



4.2 Data Visualization

- Correlation heatmap (相关热图): used to show the correlation between numerical features.





End of Chapter 4